

the data but cannot decode it if he does not know how to do so; if the reader knows how to decode the data, it isn't secure. In steganography, the reader isn't even aware that the data is being sent! It is often accepted that communications can be secured by encrypting the data, but this is not very reliable in practice. To ensure a high level of security, the ideal combination would be of encrypting the data and then sending it across using steganography.

Watermarking, as opposed to steganography, needs the additional protection against attempts by pirates at distorting/removing the watermark. In this context, the robustness depends on the application.



Copyright being the main area of watermarking, the data need not always be hidden, as some systems use visible digital watermarks. However, the media has mostly focused

Steganalysis

The art of detecting, decoding, erasing and altering messages hidden via steganography is called steganalysis. It is easiest when the before as well as the after steganography copies of the file are present. Steganalysis, can make the hidden data work against the creator. Any malicious interceptor could alter a carrier file without the knowledge of either the sender or the intended receiver. Hence, inaccurate or wrong data could be passed under the identity of the original sender.

Steganalysis tools are found aplenty on

the Net. One only needs to be aware of the carrier files; what follows is a three-step process—acquire, extract and destroy or modify. Stegdetect, for instance, is an automated tool for detecting steganographic content in images. It is capable of detecting several different steganographic methods that embed information in JPEG images. Currently, the detectable schemes are jsteg, jphide (Unix and Windows), Invisible Secrets, Outguess 1.3b, F5, AppendX and Camouflage.

on imperceptivity—invisible, inaudible digital watermarks that have wider applications.

Modern visible watermarks may be visual patterns such as a company logo or copyright sign overlaid on digital images. In most cases the information hidden using steganographic techniques is not related at all to the carrier file. These differences in goal lead to very different hiding techniques.

The final truth

Technology is a double-edged sword,

which can be used and misused depending on the user. Steganography is an example of this sword. It can be used for safeguarding data, such as in the field of media where copywriting ensures authenticity. On the flip side, many a terrorist and anti-humanist activities have been carried out cloaked under this technique.

While we may be able to put our trust in good faith and sense, predecessors of steganography, encryption and ciphering have proved that there is more harm than good to be done by it. 

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